

CHAPTER – II

IDEATION PHASE

2.1 – Purpose of the Project

Plugging into the Future: An Exploration of Electricity Consumption Patterns

Electricity demand is increasing rapidly due to population growth, urbanization, industrial expansion, and the widespread use of electronic devices and appliances. However, many consumers and organizations lack a clear understanding of how, when, and where electricity is being consumed. This limited visibility often leads to inefficient energy usage, higher electricity costs, increased strain on power grids, and greater environmental impacts resulting from excessive energy production.

Furthermore, electricity consumption patterns vary significantly across households, businesses, and industries, influenced by factors such as weather conditions, lifestyle choices, appliance efficiency, and economic activities. Without proper analysis of these patterns, it becomes difficult to identify energy wastage, predict peak demand periods, and implement effective energy management strategies.

The challenge is to analyze electricity consumption data to uncover meaningful trends and factors affecting energy use, while identifying opportunities for improving efficiency and promoting sustainable consumption practices. Understanding these patterns is essential for reducing energy costs, enhancing grid reliability, supporting renewable energy integration, and contributing to long-term environmental sustainability.

Core Problem

How can electricity consumption patterns be effectively analysed and understood to identify inefficiencies, reduce energy waste, optimize energy usage, and support sustainable energy management for the future?